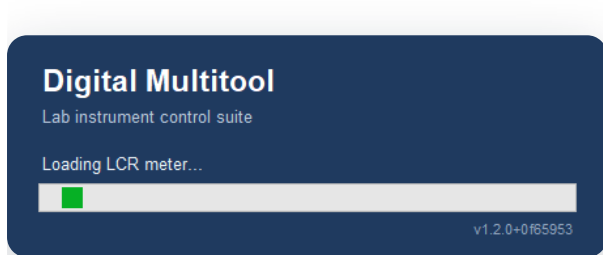


DIGITAL MULTITOOL • V1.2.0

Digital Multitool

User manual for the lab bench-control app — one window for the LCR meter, oscilloscope, signal generator, DC supply, DMM, data logging, battery post-processing, webcam captures and automated SLDEA tests.



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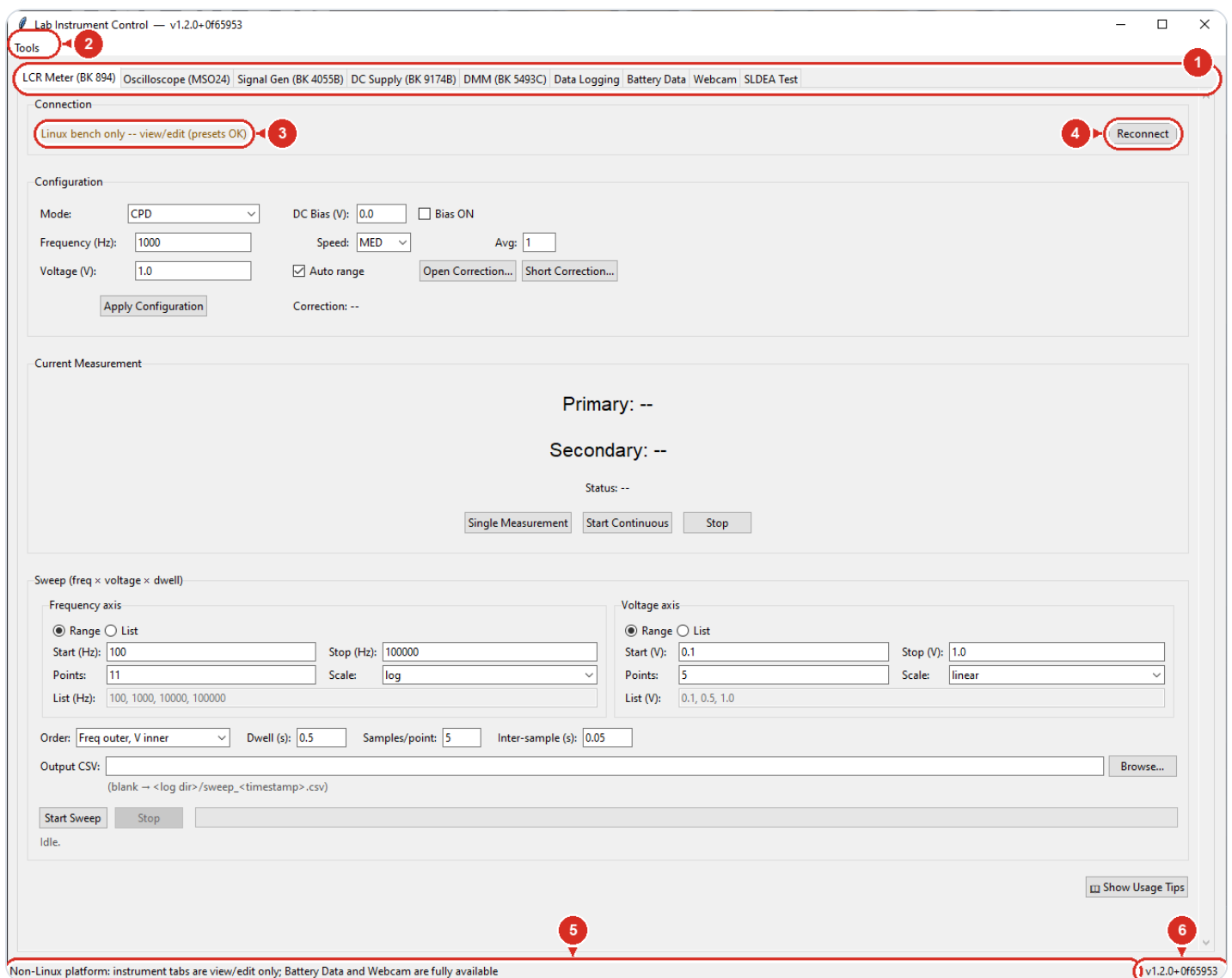
Tip: this PDF has the same chapters in the bookmarks sidebar of your PDF viewer — keep it open for one-click navigation.

Getting started

The frame around the instrument tabs: the launch splash, each tab's connection line with its one-click Reconnect, the Tools menu (whole-bench profiles and the software updater), and the footer status bar with the exact build version.

LAUNCHING

- **Bench PC (Linux):** double-click the desktop icon, or run `.venv/bin/python gui.py`.
- **Windows lab PC:** double-click `_software\Launch_SCPI_Control_Windows.bat` on the ShareDrive. First run needs Python 3.10+ and internet.
- On Windows the instrument tabs are **view/edit only** (amber note in each tab) — Battery Data and Webcam (snapshots and timelapse; not sig-gen-driven sweeps) are fully functional, and presets and bench profiles can be prepared and shared.



1 One tab per instrument or job — click to switch

2 Tools menu — bench profiles & software update

3 Connection status — green once the instrument answers (amber on Windows)

4 Re-detects this one instrument after power-on or replug

5 Status bar — confirmations and warnings appear here

6 Exact version — quote it when reporting a problem

TYPICAL USE

1. Double-click the launcher; the Digital Multitool splash names each tab as it loads.
2. The main window opens and all instruments auto-connect in the background.
3. Check each tab's Connection line turns green with the instrument's identity.
4. If a line is red, power that instrument on and click its Reconnect.
5. Tools > Load Bench Profile... to recall a saved whole-bench setup.
6. Work; watch the footer status bar for confirmations and warnings.
7. Close the window when done — sig-gen outputs and LCR bias switch off automatically; the DC supply output is left as-is.

WATCH OUT

Closing the window switches off sig-gen outputs and LCR bias and stops a running SLDEA test (best-effort, a few seconds to ramp) — but the DC supply output is never touched. End a live HV run with ■ Abort, not the close button.

Loading a bench profile immediately rewrites every connected instrument's settings — sig-gen outputs are never switched, but the LCR DC bias follows the profile's Bias ON state. Don't load one mid-measurement.

On Windows the amber "Linux bench only" connection lines and the greyed-out Update Software... are normal, not faults.

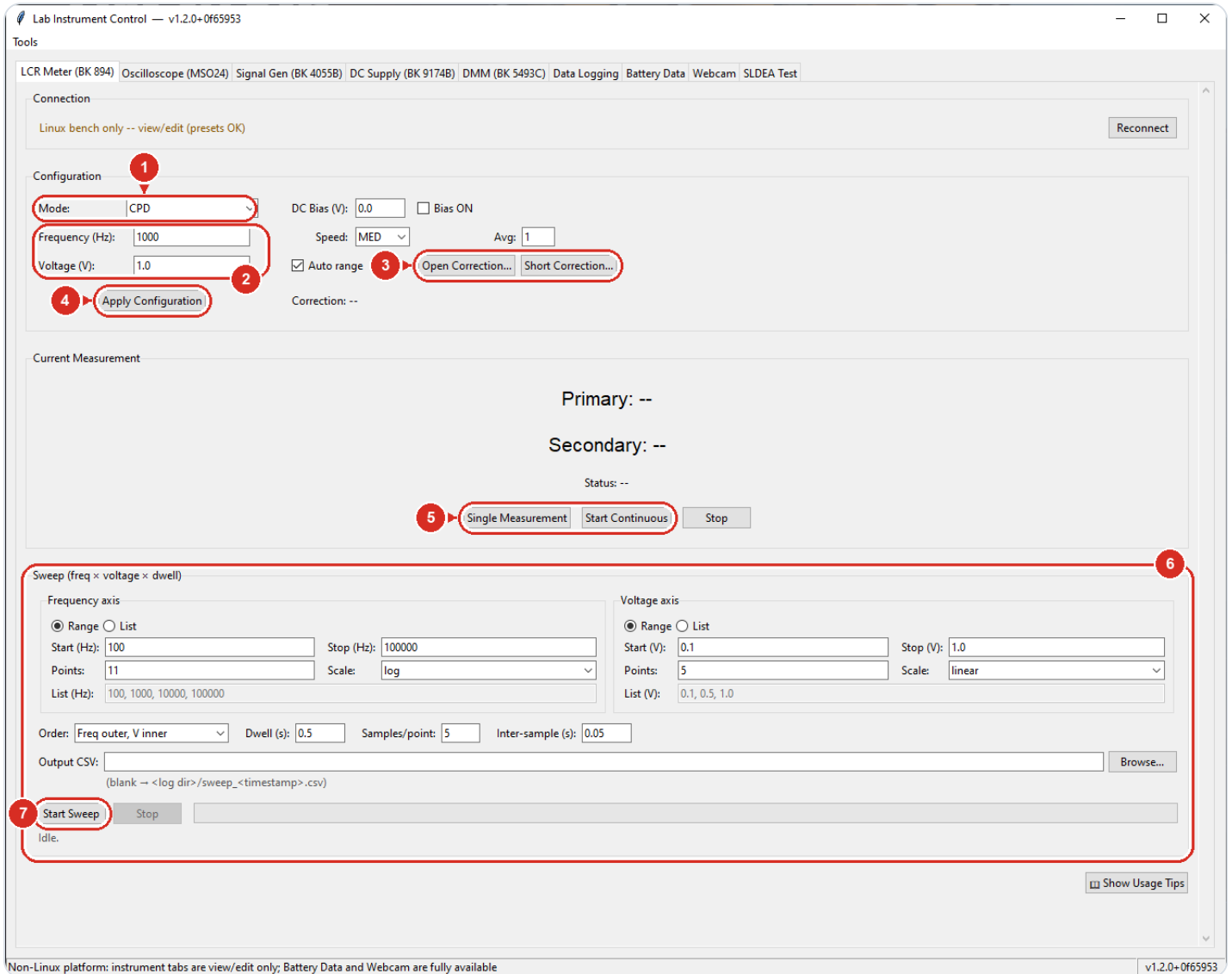
▼ The app shell, menu and profiles in detail

Splash screen (Digital Multitool)	Appears instantly at launch and names each tab as it builds (a few seconds), then closes itself.
Window title	Shows the app name plus the exact version and build hash you are running.
Connection line (top of each instrument tab)	Green 'Connected (USB/LAN/Serial)' with the instrument's identity, red 'Not connected' with the reason, amber 'Linux bench only' on Windows.
Reconnect (top-right of each instrument tab)	Closes and reopens that one instrument; use after powering a box on or replugging its cable.

Status bar (footer, left)	Running commentary: save/load confirmations, busy notices, and shared-drive fallback warnings.
Version readout (footer, right)	Version plus build hash; quote it when reporting a problem.
Tools > Save Bench Profile...	Names and stores a snapshot of the whole bench (LCR, scope, both sig-gen channels) in the shared library.
Tools > Load Bench Profile...	Restores a named snapshot into the fields and pushes it to whatever is connected; outputs are never switched.
Tools > Delete Bench Profile...	Removes a named profile after a confirmation.
Tools > Save Bench Profile to File... / Load Bench Profile from File...	Exports or imports a single profile as a .json file for a USB stick or email.
Tools > Update Software...	Pulls the latest release, redeploys the shared-drive copy for all users, then offers Restart now (Linux bench only).
Shared preset storage	Presets and profiles live in the ShareDrive presets folder so both bench users see them; if the NAS is down, saves land in a local copy and the status bar says so.

LCR Meter — BK 894

Controls the B&K Precision 894 LCR meter: set the AC test signal (mode, frequency, level, DC bias), take single or live readings of a component, and run automated frequency-by-voltage sweeps that save to a CSV file.



1 Measurement pair — CPD = capacitance + dissipation

2 AC test signal — meter clamps at 2.0 V and 500 kHz

3 Fixture zeroing — redo both after changing fixture or leads

4 Nothing reaches the meter until you click this

5 One reading now, or live readings every 0.2 s

6 Automated frequency × voltage sweep — define both axes here

7 Runs the sweep in the background → CSV file

TYPICAL USE

- 1. Check Connection is green; click Reconnect if not.
- 2. Pick Mode, then set Frequency and Voltage.
- 3. Optional: run Open then Short Correction on the empty/shorted fixture.
- 4. Click Apply Configuration.
- 5. Connect the part; click Single Measurement or Start Continuous.
- 6. Read Primary/Secondary; Status OK means a good reading.
- 7. For curves: set the sweep axes and click Start Sweep — results go to CSV.

WATCH OUT

- The app enforces the meter's limits — voltage 0.01–2.0 V, frequency 100 Hz–500 kHz; out-of-range entries are rejected with a Configuration Error on Apply.
- Nothing reaches the meter until Apply Configuration; the Primary/Secondary readouts are labeled by the last APPLIED mode, not whatever the dropdown currently shows.
- DC bias stays on until you untick Bias ON and click Apply — do not leave bias across a bias-sensitive part.
- Open/Short corrections belong to the current fixture and leads — redo both after changing either, or every reading carries the wrong offset.
- Each correction sweep takes tens of seconds and the meter is busy until it finishes.

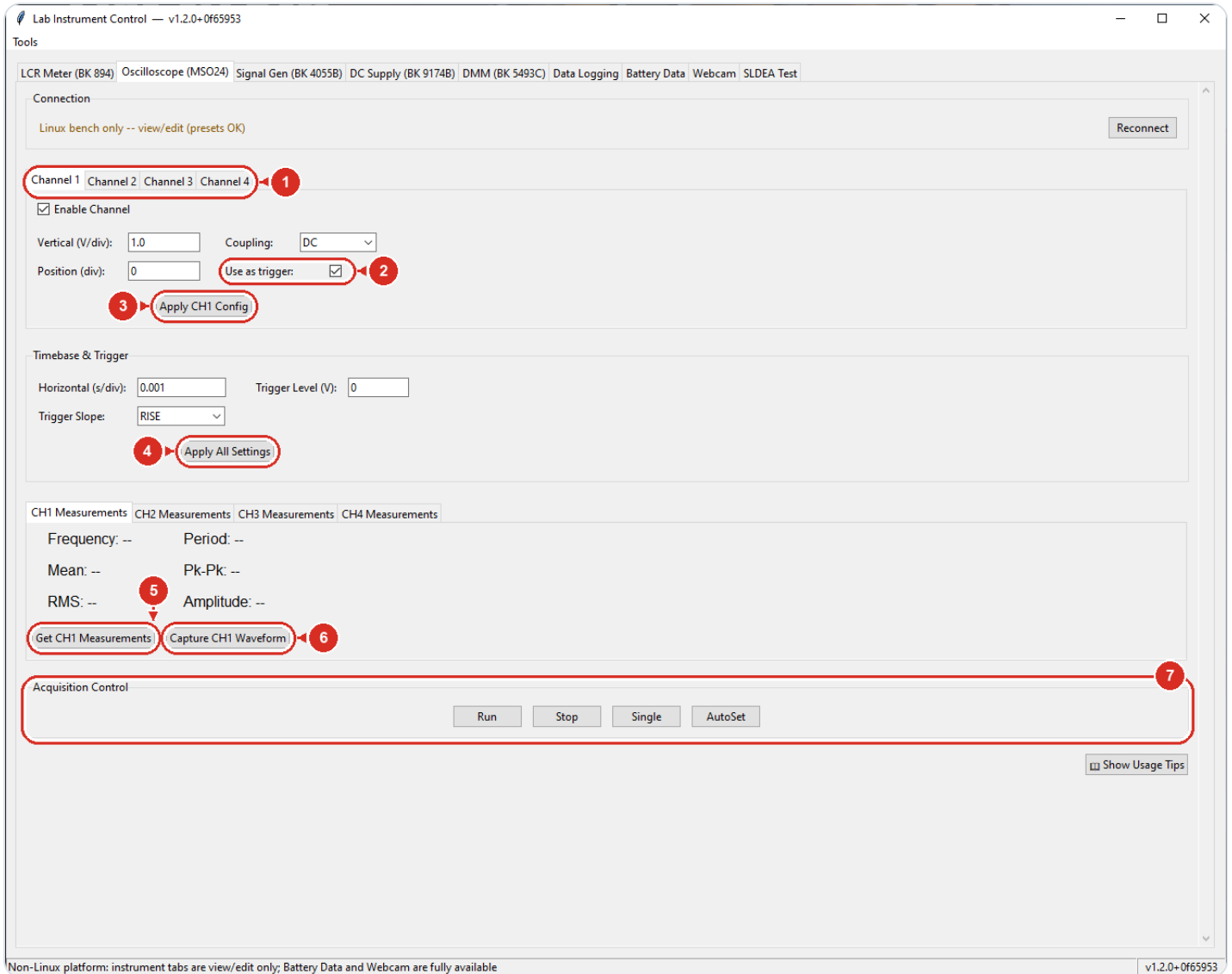
▼ Every control on this tab

Connection / Reconnect	Shows meter link state (red = not connected); Reconnect re-detects the meter and reads its current settings into the tab.
Mode	Picks the measurement pair (e.g. CPD = capacitance + dissipation); CP/LP are parallel-model, CS/LS series-model variants.
Frequency (Hz)	AC test frequency, 100 Hz to 500 kHz; 1 kHz is the standard capacitor test, 100 Hz for electrolytics.
Voltage (V)	AC test level, 0.01 to 2.0 V; 1.0 V is standard and the meter clamps anything higher to 2.0 V.
DC Bias (V) + Bias ON	Steady DC voltage held across the part during the AC test (for C-vs-bias curves); active only while Bias ON is ticked, pushed on Apply.
Speed / Avg	Reading integration time (SLOW = quietest, FAST = quickest) and how many raw readings (1-256) the meter averages into each result.

Auto range	ON lets the meter pick its range; turn OFF to hold the current range and avoid mid-sweep range-change glitches on a fixed part.
Open Correction... / Short Correction...	Fixture de-embedding sweeps (fixture empty / terminals shorted); the label beside them shows which corrections are currently ON.
Apply Configuration	Pushes everything above to the meter — no setting takes effect until this is clicked.
Current Measurement (Primary / Secondary / Status)	Latest reading, labeled by the mode actually applied to the meter; Status OK means a valid measurement.
Single Measurement / Start Continuous / Stop	One reading now, live readings every 0.2 s, or halt the live readout.
Frequency axis / Voltage axis (Range List)	Defines each sweep axis either as start-stop-points (linear or log) or as an explicit comma-separated list.
Order / Dwell (s) / Samples/point / Inter-sample (s)	Which axis is the outer loop, settling wait after each setpoint change, readings taken per point, and pause between them.
Output CSV / Browse...	Where the sweep file goes; left blank it lands in the log directory as sweep_<timestamp>.csv.
Start Sweep / Stop / progress bar	Runs the sweep in the background with live progress; Stop finishes the current sample and keeps everything written so far.

Oscilloscope — Tektronix MSO24

Controls the Tektronix MSO24 4-channel oscilloscope: set up each channel, timebase and trigger, start/stop acquisition, read automated measurements, and pull a waveform record into a plot window for saving as CSV.



1 Per-channel setup — one inner tab per channel (CH1–CH4)

2 Tick the channel that should trigger acquisition

3 Sends this channel only — does not touch the trigger

4 The only button that sends trigger source, level and slope

5 Readouts are snapshots — click to refresh

6 Pulls the trace into a plot window with Save CSV

7 Run / Stop / Single / AutoSet

TYPICAL USE

- 1. Check Connection shows connected; click Reconnect if it is red.
- 2. On each Channel tab: tick Enable Channel, set V/div, Coupling, Position.
- 3. Tick Use as trigger on the channel you want to trigger from.
- 4. Set Horizontal (s/div), Trigger Level (V) and Trigger Slope.
- 5. Click Apply All Settings (or AutoSet for an unknown signal).
- 6. Click Run, or Single for one triggered event.
- 7. On the channel's Measurements tab, click Get CHx Measurements; use Capture CHx Waveform to plot and Save CSV.

WATCH OUT

Typed values do nothing until you click an Apply button — and only Apply All Settings sends the trigger source/level/slope; Apply CHx Config does not touch the trigger.

AutoSet changes scale, timebase and trigger on the instrument but the fields on this tab do NOT update to match — clicking Apply All Settings afterwards will overwrite what AutoSet chose.

Measurement readouts are snapshots — they only update when you click Get CHx Measurements. Frequency/Period show "No signal" on non-periodic input.

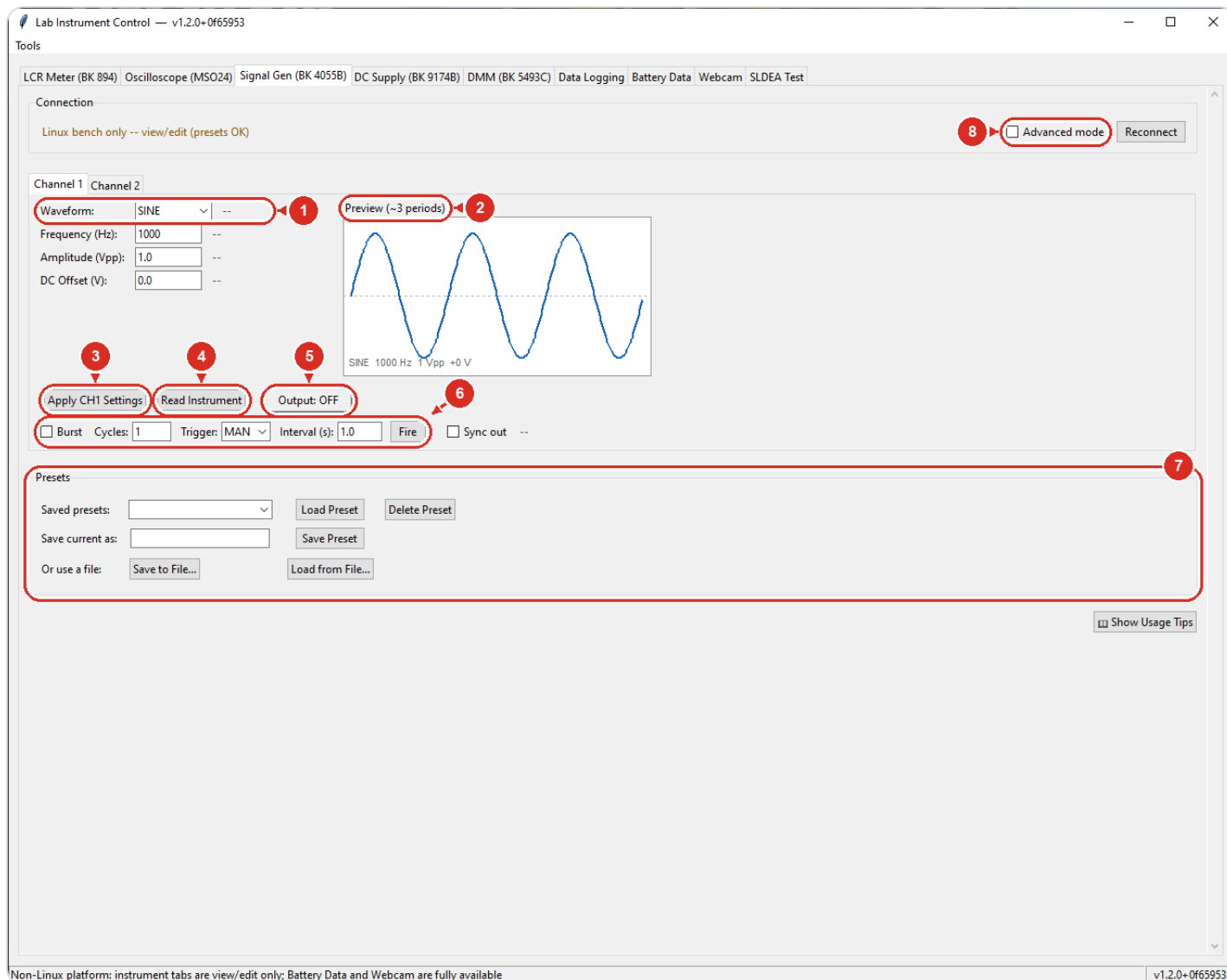
▼ Every control on this tab

Connection / Reconnect	Shows scope link status (red = not connected); Reconnect retries auto-detection over USB.
Channel 1–4 tabs	One settings page per input channel; the tab you are on is the channel you are editing.
Enable Channel	Turns the channel display on or off on the scope immediately.
Vertical (V/div)	Voltage per grid division for this channel; aim for the signal filling ~60–80% of the screen.
Coupling	DC shows everything, AC blocks the DC level (ripple on a bias), GND gives a flat zero reference.
Position (div)	Moves the trace up or down, in divisions from center.
Use as trigger	Marks this channel as the edge-trigger source; if several are ticked, the lowest number wins.
Apply CH1 Config	Sends this one channel's enable, scale, position and coupling to the scope (one such button per channel tab).
Horizontal (s/div)	Timebase — seconds per division; whole screen is 10 divisions. Show 2–3 cycles of a periodic signal.
Trigger Level (V)	Voltage the trigger-source signal must cross to start an acquisition; set it mid-signal for a stable display.
Trigger Slope	Trigger on the RISE (rising) or FALL (falling) edge.

Apply All Settings	Pushes every channel plus timebase, trigger source, level and slope in one go — the only button that sets the trigger source.
CH1–CH4 Measurements tabs	Frequency, Period, Mean, Pk-Pk, RMS and Amplitude readouts per channel; Get CHx Measurements refreshes them, Capture CHx Waveform opens the sample record in a plot window with Save CSV.
Run / Stop / Single	Run acquires continuously, Stop freezes the display, Single arms one triggered capture then stops.
AutoSet	Lets the scope pick scale and trigger for an unknown signal — good starting point, then fine-tune.

Signal Generator — BK 4055B

Controls both output channels of the BK 4055B function generator: waveform type, frequency, amplitude, offset, burst shots, and the physical output switch. Named presets let you save and recall complete two-channel setups.



1 Waveform type — the fields below adapt to it

2 Drawing of your typed settings — not a measurement

3 Pushes settings; never switches the output

4 Pull the front-panel state back into the fields

5 The real output switch — asks first, green while ON

6 Burst mode — exactly N cycles per trigger; Fire sends one

7 Named presets store both channels; files travel by USB

8 Reveals phase, pulse edges, load and polarity

TYPICAL USE

- 1. Pick the Channel 1 or Channel 2 inner tab.
- 2. Choose a Waveform; the fields adapt to it.
- 3. Enter frequency, amplitude and offset; check the preview.
- 4. Click Apply CH1 Settings and compare the gray applied readouts.
- 5. Click Output: OFF and confirm to energize the output.
- 6. For shot testing: tick Burst, set Cycles, Apply, then Fire.
- 7. Click the green Output button to switch off when done.

WATCH OUT

Turning Output ON can energize the Trek HV amplifier: 1 V from this generator = 1 kV at the DEA. The confirmation defaults to No — check amplitude and offset first.

If a gray applied readout differs from what you typed, the instrument clamped or coerced your value — believe the gray number.

Amplitude is calibrated for the selected Load (Advanced mode). With Load set to 50 Ω but the output actually open/High-Z, the real voltage is about twice what you asked for.

Custom (ARB) waveform data cannot be sent over USB — the instrument's firmware caps USB commands at 52 bytes and larger transfers freeze it until a power cycle. The app blocks such uploads; use LAN or the Waveform Editor's flash-drive .bin route instead.

▼ Every control on this tab

Reconnect	Retry the connection to the generator (tries LAN first, then USB).
Advanced mode	Reveals the extra fields: Phase, pulse Rise/Fall/Delay, Load and Polarity; Apply pushes them even when hidden.
Channel 1 / Channel 2	Inner tabs — each output is configured independently.
Waveform	Pick SINE, SQUARE, RAMP, PULSE, NOISE, DC or ARB; the fields below change to match.
Frequency / Amplitude / DC Offset (and per-waveform fields)	Type values; the gray text beside each field shows what the instrument actually accepted after Apply.
Waveform Editor...	Shown only when Waveform = ARB — opens the arbitrary-waveform designer window; the label beside it names the chosen arb.
Preview (~3 periods)	Live drawing of your typed-in settings — it is not a measurement.

Apply CH1 Settings / Apply CH2 Settings	Push the configuration to the instrument; never switches the output on or off.
Read Instrument	Pull the generator's current state back into the fields — use after changing things on the front panel.
Output: OFF / ON	Switches the real output; asks for confirmation before turning ON and goes green while ON.
Burst + Cycles / Trigger / Interval	Emit exactly N cycles per trigger, then idle — bounded energy per shot; MAN = Fire button, INT = auto-repeat, EXT = rear Aux In.
Fire	Sends one manual burst trigger (needs Burst on, trigger MAN, output ON).
Sync out	Puts a trigger edge on the rear Sync BNC every period — feed it to the scope for stable triggering.
Save Preset / Load Preset / Delete Preset	Named presets store BOTH channels; Load fills the fields and pushes the configuration to the instrument (output state untouched).
Save to File... / Load from File...	Export/import a preset as a .json file so it can travel on a USB stick or live in a project folder.

Arbitrary Waveform Editor

Design a custom waveform for the BK 4055B signal generator by placing points on a plot and choosing the shape between them. Save it to the app's library, then deliver it to the instrument by flash-drive file (.bin or EasyWaveX CSV) or by LAN upload. Opened from the Signal Gen tab: set a channel's waveform to ARB and click "Waveform Editor..."

The screenshot shows the 'Arbitrary Waveform Editor - CH1' window. At the top, a status bar displays 'Points: 4096', 'Set', 'samples (max 16384)', 'Full-scale ±V: 10', 'Set', and '4096 samples · period 1 ms = 1000 Hz · 4.1 MSa/s'. Below this is a 'Points' table with columns 'X (ms)', 'Y (V)', and 'To next'. The table contains 8 rows of data. To the right of the table is a plot area showing a waveform with points and connecting lines. Below the plot is a 'Segment after point' section with a 'Type' dropdown, an 'Advanced' checkbox, and some formulas. At the bottom, there is a 'Library & Upload' section with fields for 'Name', 'Save to Library', 'Load', 'Delete', 'Import CSV...', 'Export CSV...', 'Save Template...', 'Export .bin for 4055B flash drive...', 'Export for EasyWaveX (flash drive)...', 'Send to CH: 1', and 'Upload & Select'. Red numbered callouts (1-8) highlight specific features: 1 points to the top status bar, 2 points to the 'Points' table, 3 points to the plot area, 4 points to the 'Segment after point' section, 5 points to the 'Name' field, 6 points to the 'Export .bin for 4055B flash drive...' button, 7 points to the 'Export for EasyWaveX (flash drive)...' button, and 8 points to the 'Upload & Select' button.

X (ms)	Y (V)	To next
0.000	0.000	LINE
0.150	10.000	LINE
0.450	10.000	LINE
0.550	10.000	LINE
0.750	10.000	LINE
0.850	0.000	LINE
1.000	0.000	-

Segment after point
Type:
☐ Advanced (offset, phase, edges)
Vpk = peak (±) about the baseline.
Sine: Vpp = 2×Vpk, Vrms ≈ 0.707×Vpk.

Library & Upload
Name: Save to Library
Import CSV... Export CSV... Save Template...
Export .bin for 4055B flash drive... Export for EasyWaveX (flash drive)...
Send to CH: Upload & Select

Click canvas to add a point - drag a dot to move it - right-click a dot to delete. Edit exact X/Y in the table (double-click a cell). Ctrl+Z / Ctrl+Y to undo/redo.

1 Resolution (samples) and the volt scale (max ±10 V)

2 Breakpoint table — double-click a cell for exact values

3 Click to add a point, drag dots, right-click deletes

4 Shape to the next point: LINE, HOLD, SINE, EXP...

5 Store the waveform for reuse

6 Preferred delivery: file for the 4055B's front USB port

7 Fallback: CSV template for the EasyWaveX PC

8 Direct upload — works over LAN only

TYPICAL USE

1. On the Signal Gen tab, set the channel waveform to ARB and click "Waveform Editor..."
2. Set Points, Full-scale $\pm V$, and the Time unit.
3. Click the plot to add points and drag the dots into shape.
4. Select a point and pick the segment type (and parameters) to the next point.
5. Check the header readout — the X span is one period, which sets the output frequency.
6. Type a Name and click Save to Library.
7. Click "Export .bin for 4055B flash drive..." and save to a USB stick (or "Upload && Select" if on LAN).

WATCH OUT

The export buttons only save a file — nothing is sent to the instrument. The .bin holds the shape only; the EasyWaveX CSV embeds frequency/amp/offset in its header.

After recalling the .bin on the 4055B front panel, click Apply CHx Settings on the Signal Gen tab to push frequency, amplitude and offset (the export pre-fills them).

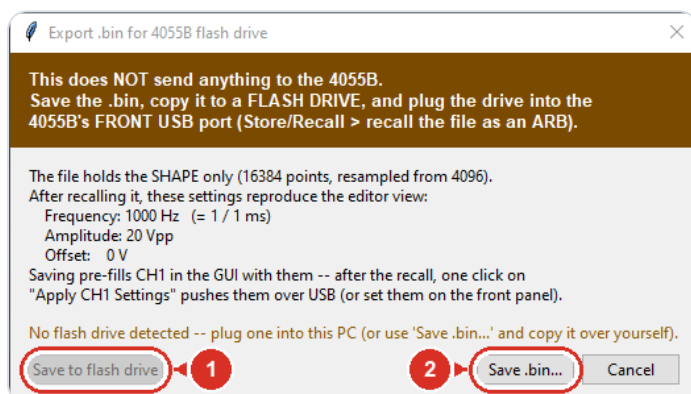
"Upload && Select" works only over a LAN connection. Over USB the upload is refused (the 4055B's 52-byte firmware cap can wedge the box) — use the flash-drive .bin route instead. With no instrument connected (e.g. the Windows viewing copy) it reports "not connected"; designing and exporting still work.

Closing the editor with unsaved edits discards them after one confirmation — anything not saved to the library is gone.

▼ Every control on this tab

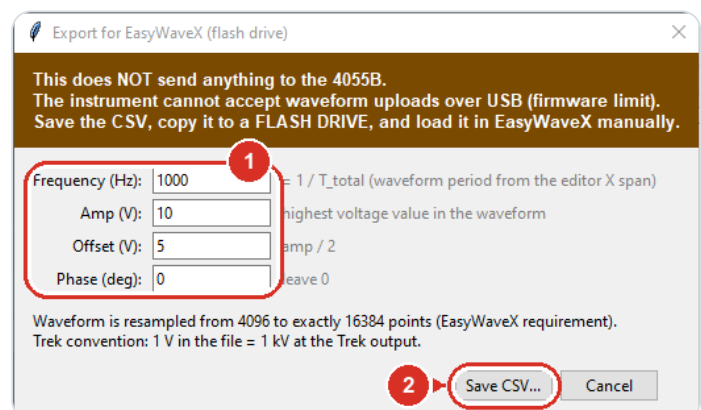
Points + Set	Total samples in the waveform (resolution), 2 to 16384.
Full-scale $\pm V$: + Set	Volts at the top/bottom of the plot; capped at ± 10 V (4055B limit).
Header readout	Live summary: samples, period (= X span), derived frequency, sample rate.
Points table	Lists every breakpoint; double-click an X or Y cell to type an exact value.
Add / Del	Insert a point mid-interval after the selected row, or delete the selected point.
Segment after point	Shape from the selected point to the next: LINE, HOLD, SINE, SQUARE, RAMP, PULSE, or EXP, with per-shape parameters.

Advanced (offset, phase, edges)	Reveals the extra segment parameters (offset, phase, rise/fall).
Preview plot	Click empty space to add a point, drag a dot to move it, right-click a dot to delete it.
Fit All / Zoom - / Zoom + / scrollbar	Zoom and pan the preview; Fit All shows the whole waveform.
Periods / Time unit	Preview 1-3 repeats; time unit (μs /ms/s) sets what X means — the X span is one period, so it sets the output frequency.
Undo / Redo	Step edits back/forward (Ctrl+Z / Ctrl+Y, 50 levels).
Name + Save to Library	Store the waveform (re-editable) in the app library under this name.
Library list + Load / Delete	Recall a saved waveform into the editor, or remove it from the library.
Import CSV... / Export CSV... / Save Template...	Bring in or write out the raw sample list; Template writes an example CSV to fill in.
Send to CH + Upload && Select	Push the waveform to the chosen channel over LAN and select it (needs a LAN-connected 4055B).



1 One click straight onto a detected USB stick

2 ...or choose any folder yourself

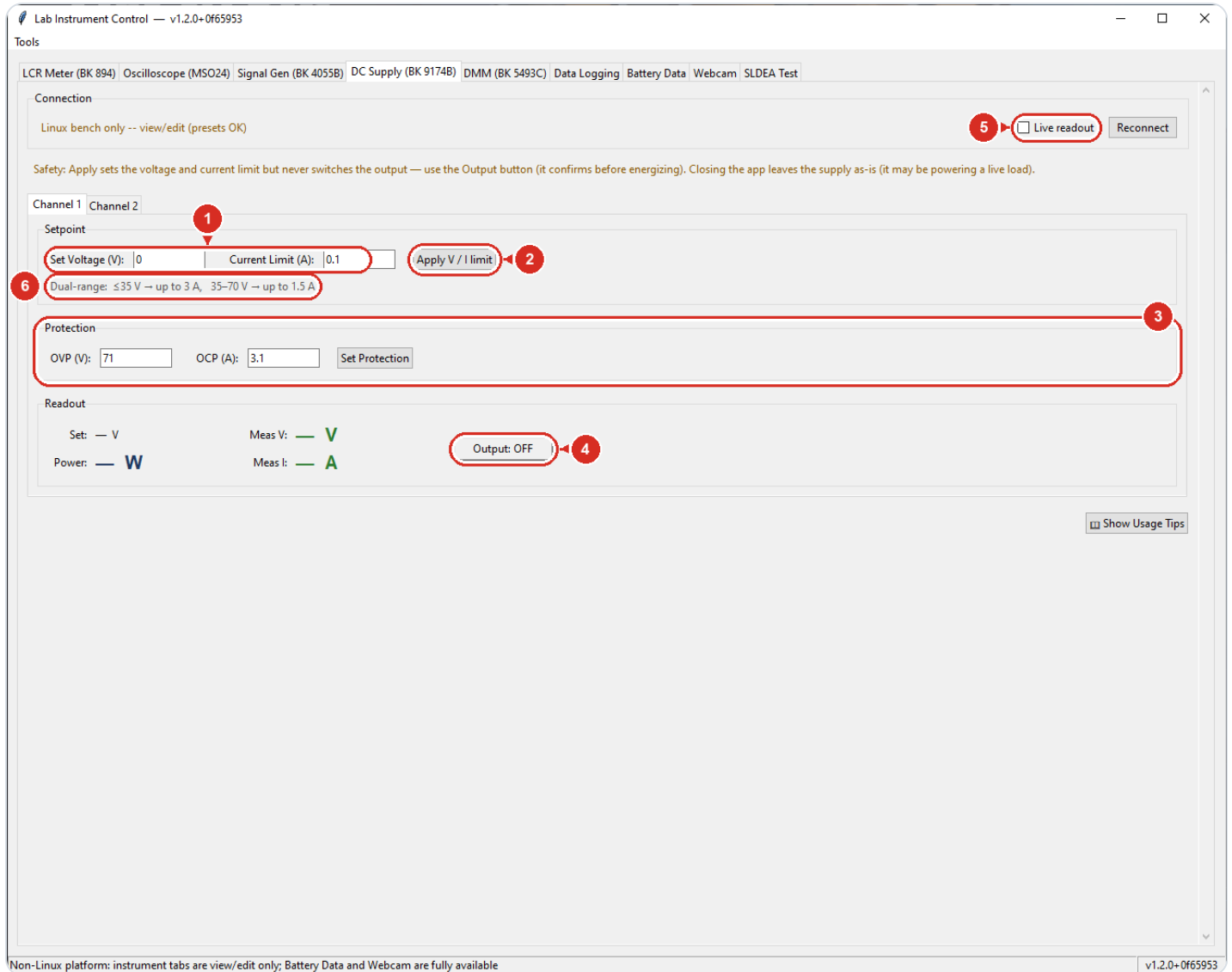


1 Header values — lab defaults pre-filled (1 V = 1 kV at the Trek)

2 Writes the exact template EasyWaveX expects

DC Supply — BK 9174B

Controls the B&K 9174B dual-output bench DC supply: per channel you stage a voltage and current limit, set hardware protection trips, watch live measured volts/amps/watts, and switch the output on or off. Setting values and energizing the terminals are deliberately separate actions.



1 Stage voltage and current ceiling (CC — not a fuse)

2 Sends values — output stays untouched

3 OVP / OCP hardware trips — set these first

4 Energizes the terminals — confirms first, green while ON

5 Polls measured V, A and W every 0.5 s

6 Allowed envelope: 3 A up to 35 V, 1.5 A above

TYPICAL USE

1. Check Connection is not red; press Reconnect if needed.
2. Pick the Channel 1 or Channel 2 sub-tab.
3. Enter OVP and OCP, press Set Protection.
4. Enter Set Voltage and Current Limit, press Apply V / I limit.
5. Press the Output button and confirm — it turns green ON.
6. Tick Live readout to watch measured V, A, and W.
7. Press the green Output button again to switch off when done.

WATCH OUT

Apply V / I limit never switches the output — only the Output button energizes the terminals, and it asks first.

Closing the app leaves the supply exactly as-is: an ON output keeps powering the load.

The current limit is constant-current, not a fuse — at the limit the supply lowers voltage to hold the current, so set it just above the expected draw.

Above 35 V each output is capped at 1.5 A; Apply refuses voltage/current pairs outside the dual-range envelope.

▼ Every control on this tab

Connection / Reconnect	Shows link status (red = not connected); Reconnect reopens the supply's USB-serial port.
Live readout (checkbox)	Polls measured voltage, current, and power on both channels every 0.5 s (read-only).
Channel 1 / Channel 2 (sub-tabs)	Pick which of the two independent outputs the panel below controls.
Set Voltage (V)	Voltage to program on this channel (0–70 V); takes effect on Apply.
Current Limit (A)	Constant-current ceiling for this channel; the supply drops voltage to hold it.
Apply V / I limit	Sends voltage and current limit to the channel — never touches the output switch.
Dual-range note	Allowed combinations: up to 3 A at ≤ 35 V, only 1.5 A from 35–70 V; Apply refuses values outside this.
OVP (V) / OCP (A)	Over-voltage / over-current hardware trip points (defaults 71 V / 3.1 A).
Set Protection	Writes the OVP/OCP trip points to the channel; do this before enabling output.
Readout (Set / Meas V / Meas I / Power)	Programmed voltage plus measured volts, amps, and computed watts; updates with Live readout on.

Output: OFF / ON (button)

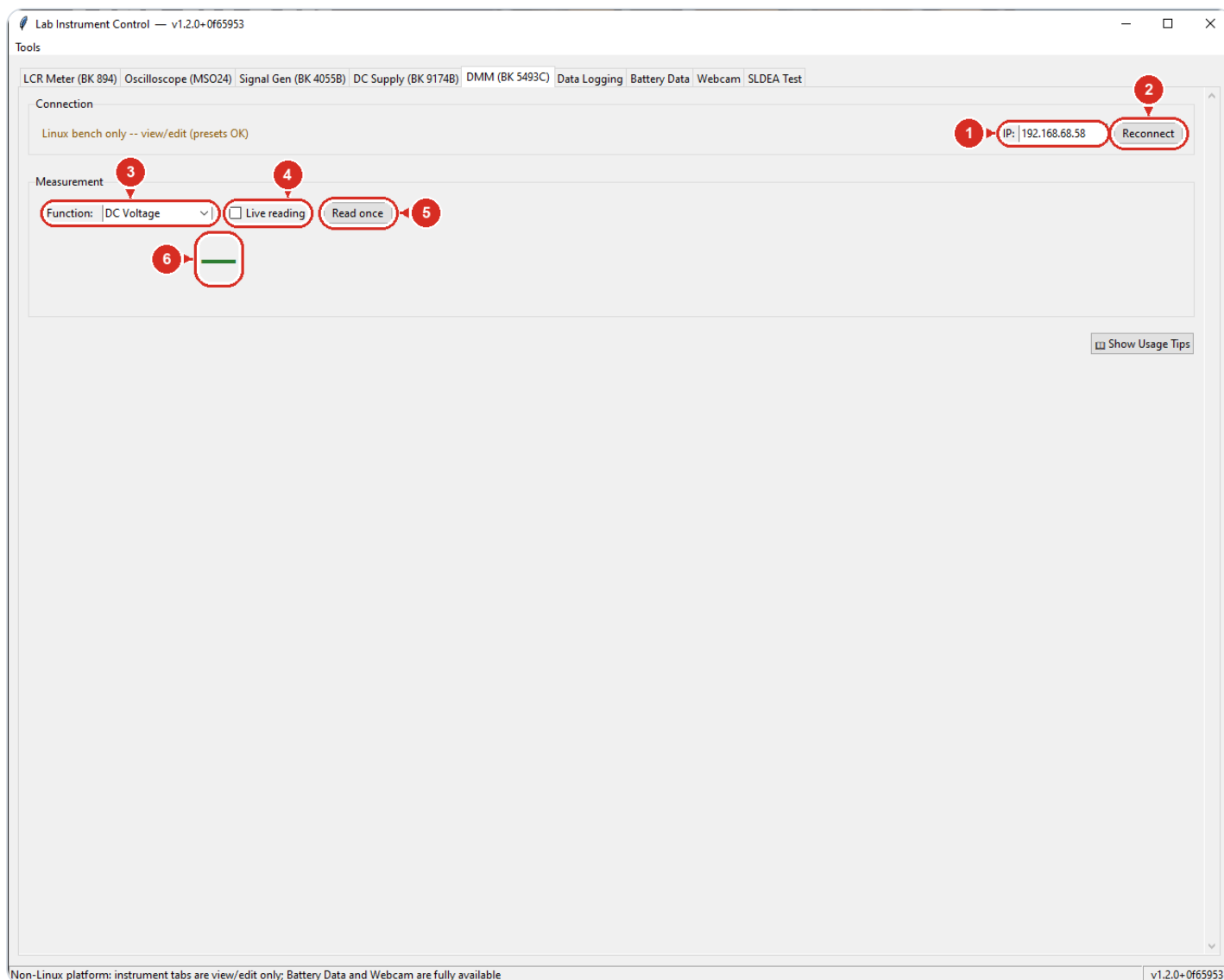
Toggles the channel output; asks for confirmation before energizing and turns green when ON.

 **Show Usage Tips**

Pops up a quick-reference summary of this tab.

Digital Multimeter — BK 5493C

Reads the bench B&K 5493C 6.5-digit multimeter over the lab network (this unit is Ethernet-only). Pick a measurement function and take a single reading or watch a live, continuously updated display.



1 The meter's LAN address — this unit is Ethernet-only

2 Connect / retry at this IP

3 What to measure — V, A, Ω , Hz, F (auto-ranged)

4 Continuous display, ~2 readings per second

5 Single measurement

6 Reading appears here — red OVLD on overload

TYPICAL USE

- 1. On the meter's front panel: Menu -> I/O / LAN, use "DHCP Once" to get an address on the bench subnet.
- 2. Type that IP into the IP: field and click Reconnect; status turns green.
- 3. Pick a Function from the dropdown (starts on DC Voltage).
- 4. Click Read once, or tick Live reading for a continuous ~2 Hz display.
- 5. Read the big number; the grey line beneath shows the raw value and unit.
- 6. To record: tick "DMM" on the Data Logging tab — it logs the function selected here to CSV.

WATCH OUT

This unit is LAN-only — its USB never enumerated (known hardware issue). Do not troubleshoot USB; fix the network instead.

The meter's socket accepts one client at a time. If Reconnect keeps failing, close any other program (or stale session) talking to it, or power-cycle the meter.

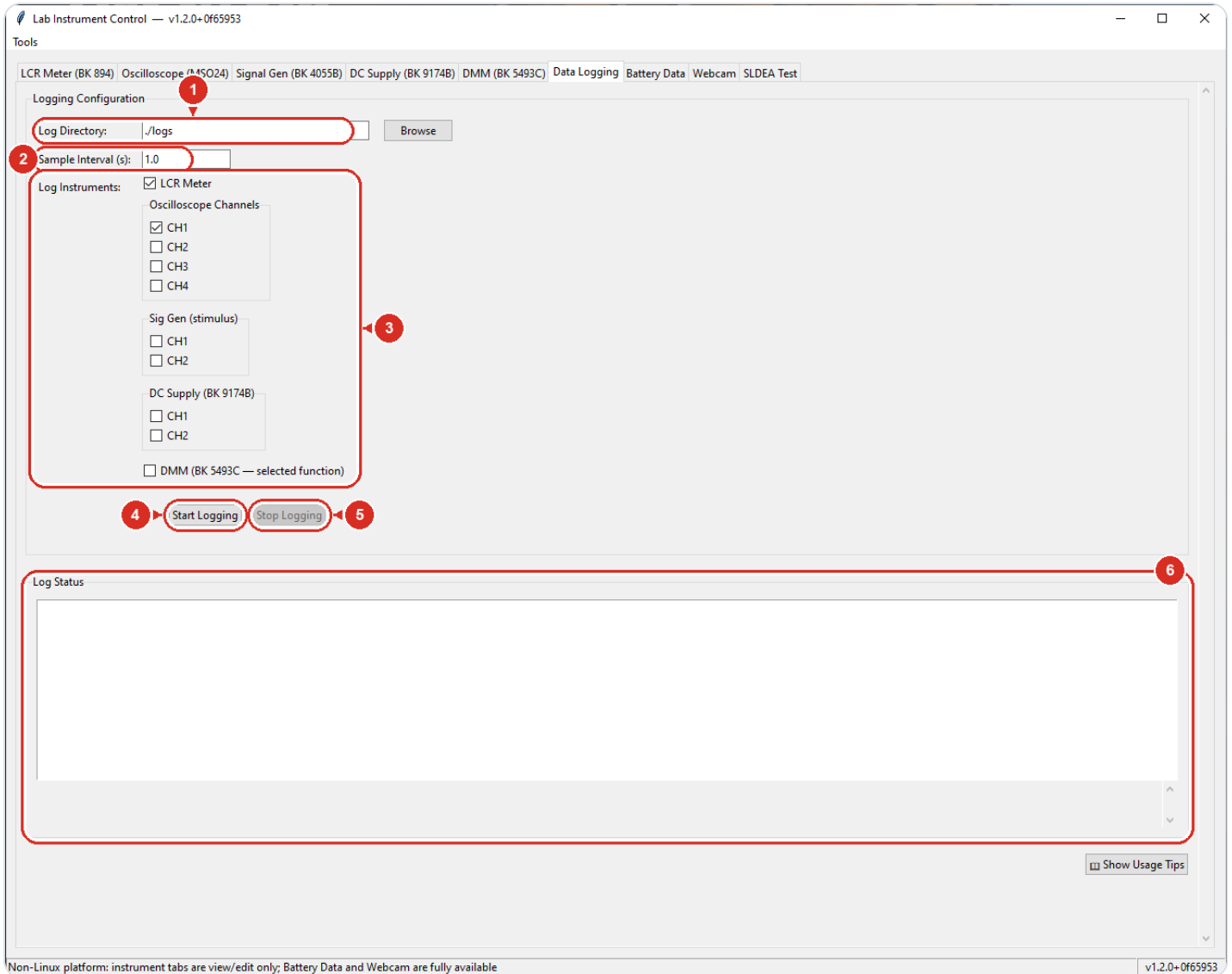
Data Logging records the Function selected at the moment you start logging; changing the dropdown mid-log does not change what is recorded.

▼ Every control on this tab

Not connected (status)	Shows link state to the meter; red until a connection succeeds.
IP:	The meter's LAN address (default 192.168.68.58); set it on the meter's front panel first.
Reconnect	Connects (or retries) to the meter at the IP shown, on port 45454.
Function:	What to measure: DC/AC Voltage, DC/AC Current, 2W/4W Resistance, Frequency, Capacitance — all auto-ranged.
Live reading	Polls the selected function every 0.5 s (~2 Hz) while ticked.
Read once	Takes a single measurement of the selected function.
Big readout	Latest reading in engineering notation (e.g. 16.74 mV); turns red "OVLD" on overload or a non-numeric reply.
Small line under readout	The function name plus the raw value and unit.
☐ Show Usage Tips	Opens a short reference on connection, use, and logging.

Data Logging

Records readings from the connected instruments to CSV files at a fixed interval — one timestamped file per source, so a run captures stimulus (sig gen, supply) and response (LCR, scope, DMM) side by side. Files are for offline analysis (Excel, Python); there is no live plot on this tab.



1 All CSV files land here (default ./logs)

2 Seconds between samples

3 Tick every source to record — one CSV per source

4 Opens fresh timestamped CSVs and starts sampling

5 Closes the files and ends the run

6 File paths, errors, and dropped sources appear here

TYPICAL USE

- 1. Connect and configure the instruments on their own tabs first.
- 2. Pick a Log Directory (Browse) or keep ./logs.
- 3. Enter the Sample Interval in seconds (e.g. 1.0).
- 4. Tick every source you want recorded.
- 5. Click Start Logging and note the CSV paths in Log Status.
- 6. Watch Log Status during the run for errors or dropped sources.
- 7. Click Stop Logging, then open the CSVs in Excel or Python.

WATCH OUT

A source that errors 5 times in a row is silently dropped for the rest of the run (one notice in Log Status) — check Log Status after long runs; if every source dies, logging stops itself.

The DMM is logged with whatever function was selected on the DMM tab at the moment you pressed Start Logging — changing the function mid-run does not change what is recorded.

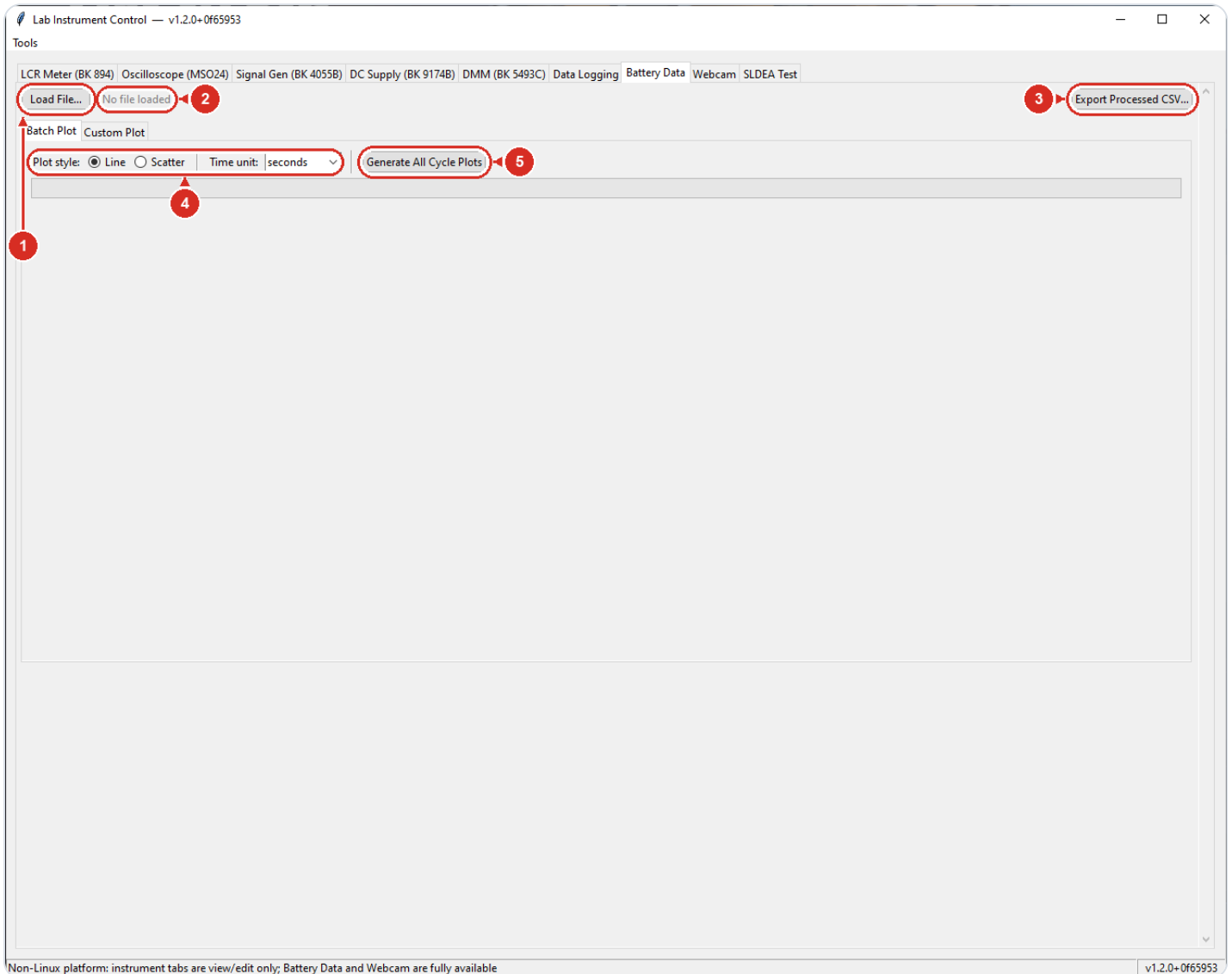
Sources are read one after another each tick, so with many sources or slow instruments the real spacing can be longer than the Sample Interval you typed.

▼ Every control on this tab

Log Directory: + Browse	Folder where the CSV files are written (default ./logs); created automatically if missing.
Sample Interval (s):	Seconds between samples; every ticked source is read once per interval.
LCR Meter	Log the LCR reading each tick: mode, frequency, primary, secondary, status.
Oscilloscope Channels (CH1–CH4)	Log scope measurements per ticked channel: frequency, period, mean, pk-pk, RMS, amplitude.
Sig Gen (stimulus) (CH1/CH2)	Log what was driving the DUT: waveform, frequency, amplitude, offset, output ON/OFF.
DC Supply (BK 9174B) (CH1/CH2)	Log set voltage, measured voltage, measured current, and calculated power per channel.
DMM (BK 5493C — selected function)	Log the function currently selected on the DMM tab (value + unit).
Start Logging	Opens one fresh timestamped CSV per ticked source and begins sampling; sources not connected are skipped with a notice.
Stop Logging	Closes the files and ends the run (takes effect within about a second).
Log Status	Run messages: where each CSV was opened, per-source errors, dropped sources.

Battery Data

Post-processes exports from the battery cycler: load the tester's .xls/.xlsx file and the tab translates the Mandarin headers and status labels to English, merges all detail sheets into one table, and converts mV to V. From there you can export a clean CSV or make plots. Works on any PC — no instrument connection involved.



1 Open the raw cycler .xls/.xlsx export

2 Shows filename + rows × columns when processed

3 Translated, merged table → CSV

4 Batch options — line/scatter, time in s/min/h

5 Two 300-dpi PNGs per cycle (V–t and V–Q)

TYPICAL USE

- 1. Click Load File... and pick the untouched cyclers export (.xls/.xlsx).
- 2. Wait until the label shows the filename with row and column counts.
- 3. Optional: click Export Processed CSV... to save the translated table.
- 4. For per-cycle PNGs: on Batch Plot, set style and time unit, click Generate All Cycle Plots, pick a folder.
- 5. For one specific plot: on Custom Plot, choose X and Y columns, style, and cycles.
- 6. Click Preview Plot to check it in the panel.
- 7. Click Save as PNG (300 dpi)... to export it.

WATCH OUT

Batch plotting overwrites existing PNGs with the same cycle-numbered names in the chosen folder without asking.

Batch plots deliberately leave out rest periods; only charge/discharge segments appear. Use Custom Plot with Color by status to see rests.

Load only the raw tester export: it must keep its original sheet layout (Info first, Cycle summary last). Re-saved or trimmed files fail to load or pull in the wrong sheets.

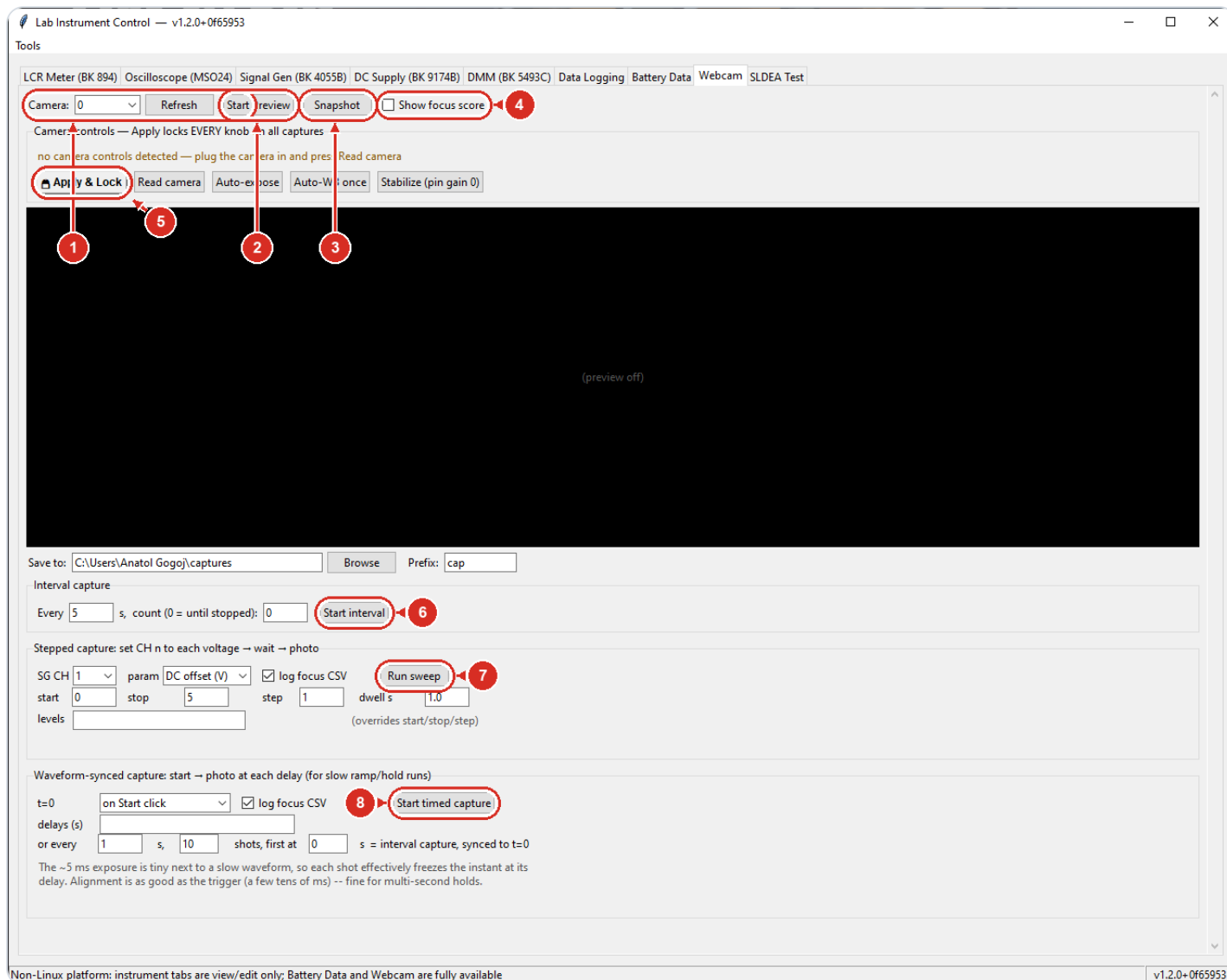
▼ Every control on this tab

Load File...	Opens a cycler .xls/.xlsx export and processes it in the background; the gray label beside it shows the filename plus row and column counts when done.
Export Processed CSV...	Saves the full translated table as a CSV (default name: <file>_processed.csv).
Batch Plot / Custom Plot	Inner tabs that switch between one-click per-cycle plotting and a build-your-own plot panel.
Plot style: Line / Scatter (Batch Plot)	Draw batch plots as connected lines or as dots.
Time unit (Batch Plot)	Seconds, minutes, or hours for the time axis of batch plots.
Generate All Cycle Plots	Asks for an output folder, then saves two 300-dpi PNGs per cycle: voltage vs time and voltage vs capacity.
Progress bar (Batch Plot)	Fills as batch plots are written; the status text ends with a count and the folder.
X Axis / Y Axis (Custom Plot)	Pick any numeric column for each axis; defaults to relative_time_s vs voltage_V.
Style: Line / Scatter (Custom Plot)	Line or dot rendering for the custom plot.
Color by status	Splits the trace into colored segments per charge/discharge/rest state, with a legend.
Time unit (Custom Plot)	Rescales time-based axes to seconds, minutes, or hours.

Cycles: All cycles / Selected:	Plot everything, or type cycle numbers like 1,2,3 or 1-5 in the entry box.
Preview Plot	Draws the plot in the panel on the right; use the toolbar under it to zoom and pan.
Save as PNG (300 dpi)...	Redraws the custom plot and saves it as a 300-dpi PNG.

Webcam

Live view and photo capture from the bench USB/industrial camera. Take single snapshots, timelapse series, or automated runs where each photo is tied to a signal-generator voltage or to a delay after a start trigger — with an on-screen focus score to help you sharpen the lens first.



1 Pick the camera; Refresh rescans devices

2 Live view — the same button stops it

3 Save one timestamped PNG

4 Sharpness number — turn the lens until it peaks

5 Locks every camera knob — prevents drift during runs

6 Automatic photo every N seconds

7 Steps a sig-gen voltage, one photo per level

8 Photos at chosen delays after t=0

TYPICAL USE

1. Pick the camera number and click Start Preview.
2. Tick Show focus score; turn the lens until the number peaks.
3. Set exposure/colour (or use Stabilize), then  Apply & Lock.
4. Set the save folder and filename prefix.
5. Snapshot for one photo, or Start interval for a timelapse.
6. For voltage-vs-image runs: fill start/stop/step and dwell, click Run sweep.
7. Each filename carries shot number, timestamp, and the level.

WATCH OUT

Before a sweep, turn the channel's Output ON in the Signal Gen tab — otherwise the levels are written but never reach the device, and you get a folder of identical photos.

A sweep leaves the generator sitting at the LAST level when it ends — it does not return the output to zero.

Without  Apply & Lock the camera's firmware auto-gain/auto-WB drifts brightness and colour mid-run.

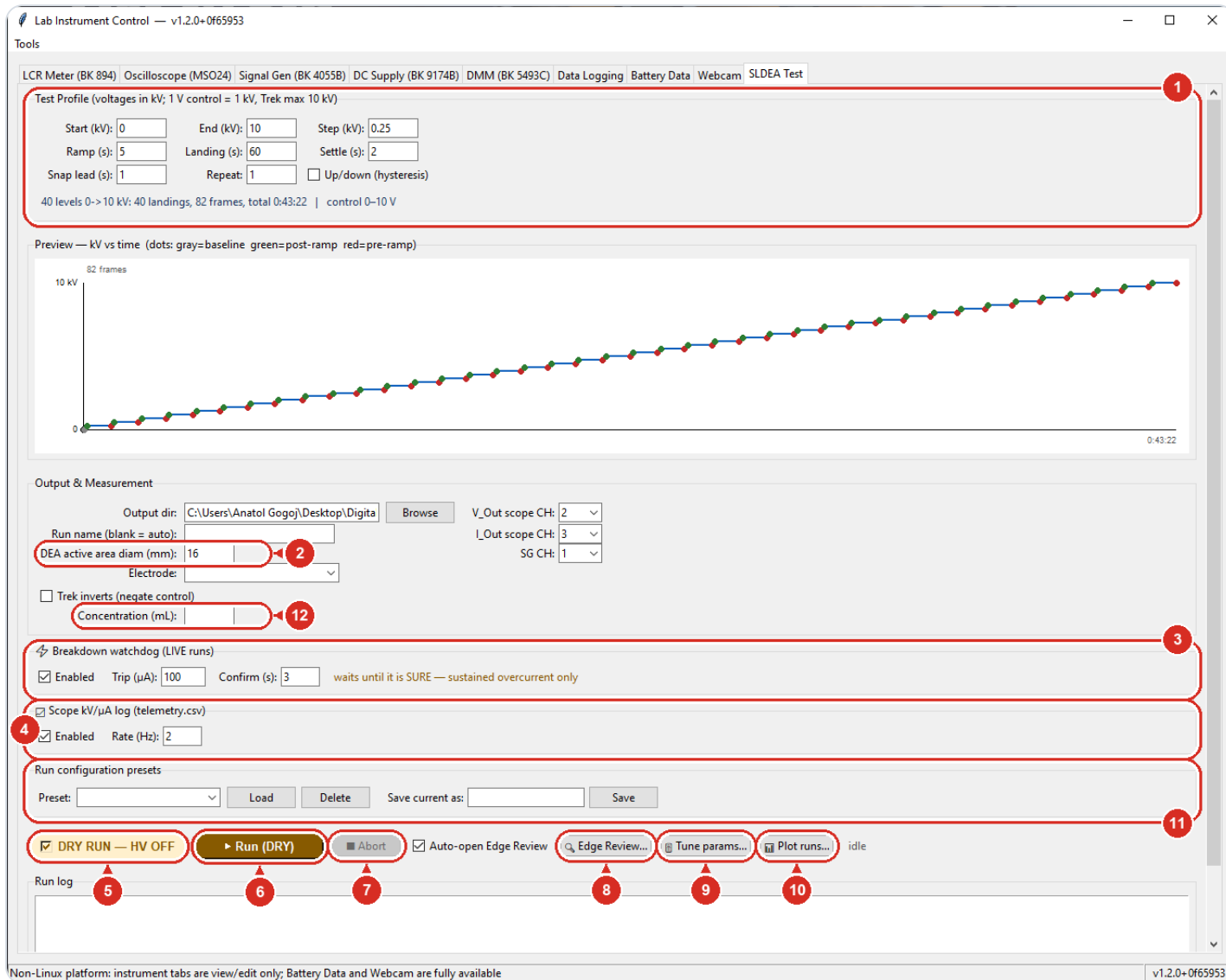
▼ Every control on this tab

Camera + Refresh	Pick the video device number; Refresh rescans after plugging or unplugging a camera.
Start Preview	Live view of the selected camera; the same button stops it.
Snapshot	Saves the current preview frame as a timestamped PNG in the save folder.
Show focus score	Overlays a live sharpness number and a green area-of-interest circle — higher = sharper; turn the lens to maximise it.
Camera controls panel	One field per camera knob (exposure, gain, white balance...), built from what the camera reports.
 Apply & Lock	Writes every panel value to the camera, keeps re-asserting them during all captures, and saves them for the next start.
Read camera	Refills the fields from the camera's current state.
Auto-expose	Searches for an exposure giving a mid-grey image and fills it in; then Apply & Lock.
Auto-WB once	One-shot grey-world colour balance on the current scene; stop the preview first.
Stabilize (pin gain 0)	Pins gain at 0 (defeats the firmware auto-gain) and finds a matching exposure — the only drift-free combo.
Save to / Prefix	Output folder and the start of every saved filename (e.g. cap_0003_20260728-141230_1p9V.png).
Interval capture	Automatic snapshot every N seconds; count 0 keeps going until you press Stop.

Stepped capture (Run sweep)	Sets the chosen sig-gen channel to each voltage (start/stop/step or an explicit list), waits the dwell, saves a photo per level.
log focus CSV	Also writes a CSV with one row per shot: level or delay, filename, sharpness score.
Waveform-synced capture	Photos at chosen delays after t=0 (Start click or a channel burst trigger) — for imaging fixed points of a slow ramp/hold run.

SLDEA Test

Runs an automated voltage-staircase test on a single-layer dielectric elastomer actuator (SLDEA): the signal generator drives the Trek HV amplifier (1 V control = 1 kV, max 10 kV), the scope logs the Trek's V_Out/I_Out monitors, and the webcam photographs the electrode at each voltage landing. Each run produces a self-contained folder (setup.txt, data.csv, run.log, frames/ — plus telemetry.csv when the scope is connected) ready for the Edge Review analysis program.



1 The voltage staircase — start/end/step, ramp and landing times (the 0 kV reference photo is always taken)

2 Sets the px→mm scale — a wrong value corrupts all areas

3 Aborts on sustained overcurrent — leave Enabled

4 Logs kV/μA continuously to telemetry.csv — the current between photos

5 Safety toggle — untick only for a live HV run

6 Starts the run — the label shows the mode

- 7 Ramps to 0 kV first, then stops
- 8 Open Edge Review on a finished run — see the Edge Review section
- 9 Advanced — confirmation-gated; Save rewrites the run's detection settings
- 10 Several finished runs on one figure — Export writes the PNG and its tidy CSV together
- 11 Save the whole tab under a name and recall it — never the run name, and never the LIVE state
- 12 How much CNT ink went on — greyed out for carbon black and eGaln, which are not inks

TYPICAL USE

1. Enter the staircase: Start/End/Step kV, Ramp and Landing times.
2. Check the preview and summary line — dots are photos, total time shown.
3. Set Output dir, DEA active area diam (mm), and the scope/SG channels.
4. Do a dry run first to rehearse timing and camera captures.
5. Untick DRY RUN; answer the scope-monitor and Energize HV prompts.
6. In camera pre-flight, center the DEA in the circle, check focus, start.
7. Let it finish (or ■ Abort); Edge Review opens on the new run folder.

WATCH OUT

A live run drives real HV up to 10 kV through the Trek. Stop with ■ Abort (it ramps to 0 kV first); closing the window also stops the run best-effort — if the SG link is dead the Trek can stay energized and the app raises an HV NOT ZEROED alarm.

If the scope-monitor check warns and you pick 'run anyway', the run completes but logs blank measured kV and wrong currents — this silently ruined five runs on 2026-07-25.

A wrong DEA diameter corrupts the px-to-mm scale: every area number from that run is wrong and setup.txt must be hand-fixed before re-analysis.

🔧 Tune params... is an advanced tool: it asks for confirmation first, and its Save rewrites the run's setup.txt — the detection settings every later analysis pass will use.

▼ Every control on this tab

Start / End / Step (kV)	Defines the voltage staircase; End must be 10 kV or below.
Ramp (s) / Landing (s)	Transition time between levels and hold time at each level.
Settle (s) / Snap lead (s)	When the two photos per landing are taken: settle = wait after the ramp before the post-ramp photo; snap lead = how long before the next ramp the pre-ramp photo fires.
Repeat / Up/down (hysteresis)	Repeat the whole sweep N times; Up/down adds a descending leg for hysteresis curves.

0 kV baseline photo	Captured automatically at the start of every run — Edge Review measures all areas against it. (No longer a checkbox: unticking it silently broke every outline.)
Preview — kV vs time	Live plot of the staircase; dots mark photo moments (gray baseline, green post-ramp, red pre-ramp) and a red playhead tracks a running test.
Output dir / Run name (blank = auto)	Where the run folder is created; blank name auto-stamps SLDEA_YYYYMMDD_HHMMSS.
V_Out / I_Out scope CH, SG CH	Which scope channels carry the Trek monitor outputs and which SG channel drives the Trek.
DEA active area diam (mm)	Nominal resting diameter of the active area — written to setup.txt and used as the px-to-mm scale in Edge Review.
Trek inverts (negate control)	Tick when the Trek is configured inverting; the run then drives negative control so the HV output stays positive.
Electrode	Which compliant electrode material this device uses. Pick one of the listed inks — Carbon Solutions P2/P3-SWNT, nano-c Invisicon 3500/3900, plain CNT, carbon black, eGaln — or type any material straight into the box; whatever you type is recorded word for word in setup.txt, alongside a canonical family so runs can be grouped later. Optional, but the run asks before starting without it. Leave it blank if the sample is genuinely unknown, rather than naming something you are not sure of.
Concentration (mL)	How much CNT ink went on this device — the '2.5mL' in a folder name like P3_2.5mL_Triazole, now recorded in the run itself. It is greyed out for carbon black and eGaln, which are not inks dispensed by volume: when it is greyed the run neither asks for it nor writes it to setup.txt, so a carbon-black run carries no ink key at all. Where it does apply, the run asks before starting without it, and only a positive number is accepted.
 Breakdown watchdog (LIVE runs)	Watches Trek current during live runs; current above Trip (μA) sustained for Confirm (s) snapshots the breakdown, ramps to 0 and aborts. Transient spikes never trip it.
 Scope kV/ μA log (telemetry.csv)	Writes the scope's kV/ μA continuously to telemetry.csv beside data.csv, so the current BETWEEN photos is recorded and a breakdown can be dated electrically. Rate (Hz) is capped at 2 — at that setting the current rows are free, because they are the breakdown watchdog's own readings, which used to be thrown away. Voltage is read less often (at most once a second) because it costs a second scope query; the commanded voltage is on every row regardless. Needs the scope; works on dry runs too. Nothing reads the file — data.csv is still the run, and Edge Review, the tuner and the diagnostic are unaffected.

Run configuration presets	Saves everything above under a name and recalls it later, so a campaign's staircase is not retyped each session. A preset covers the staircase, Output dir, scope/SG channels, DEA diameter, electrode, Trek inverts, and the watchdog and telemetry settings. It deliberately does NOT carry your run name, and never the DRY/LIVE state — loading one always leaves the tab in DRY, so arming the HV stays a separate deliberate click. Presets live in the same shared presets/library as the signal-generator presets. If a preset was written by an older or newer version and something in it no longer fits, the load says what it could not apply instead of skipping it quietly.
DRY RUN — HV OFF / ► Run / ■ Abort	Dry run rehearses timing and photos with HV off; unticking turns the row red for a live run. Abort ramps to 0 kV before stopping.
Auto-open Edge Review	When a run finishes with frames, launches Edge Review on it and starts a default detection pass.
 Edge Review... /  Tune params...	Open the analysis program, or the live slider tuner. Tune params is advanced and confirmation-gated — its Save rewrites the run's setup.txt.
 Plot runs...	Opens the plot window on several finished runs at once: tick the runs, choose area / current / power, toggle what is drawn (pre/post separately, mean line, uncertainty bands, breakdown marks) and the figure redraws as you go. Nothing is written until Export, which saves the 300 dpi PNG together with the tidy per-snapshot CSV behind it — the CSV is the figure's evidence, so the two always travel as a pair. Area mode needs runs Edge Review has saved areas for (✓ processed in the list); current and power work on raw runs, so a run can be looked at electrically before anyone has reviewed a frame. Hover the uncertainty-bands box to see where those numbers come from: they are the calibrated error budget in SLDEA_MEASUREMENT.md — $\pm 2\%$ for machine-measured levels, $\pm 1\%$ for hand-traced ones, a level mixing the two keeping the machine band — and Edge Review's conf score is a review-ordering number, never an uncertainty.
Run log	Timestamped record of the run: directory path, each landing, watchdog status, warnings.

SLDEA companion tools

Four stand-alone programs for recorded SLDEA runs — they touch no instruments, so any PC with a copy of the run data works.

EDGE TUNER

Drag sliders, watch the outlines redraw live on baseline / mid-run / late frames.  **Save** writes the values into that run's `setup.txt`. Open it from the SLDEA tab ( **Tune params...**) or double-click `Tune_SLDEA_Windows.bat` — or drag a run folder onto it.

EDGE REVIEW

Traces every frame of a run, queues the uncertain ones for a human pick, then writes the accepted areas back into the run's `data.csv`. Full walkthrough below.

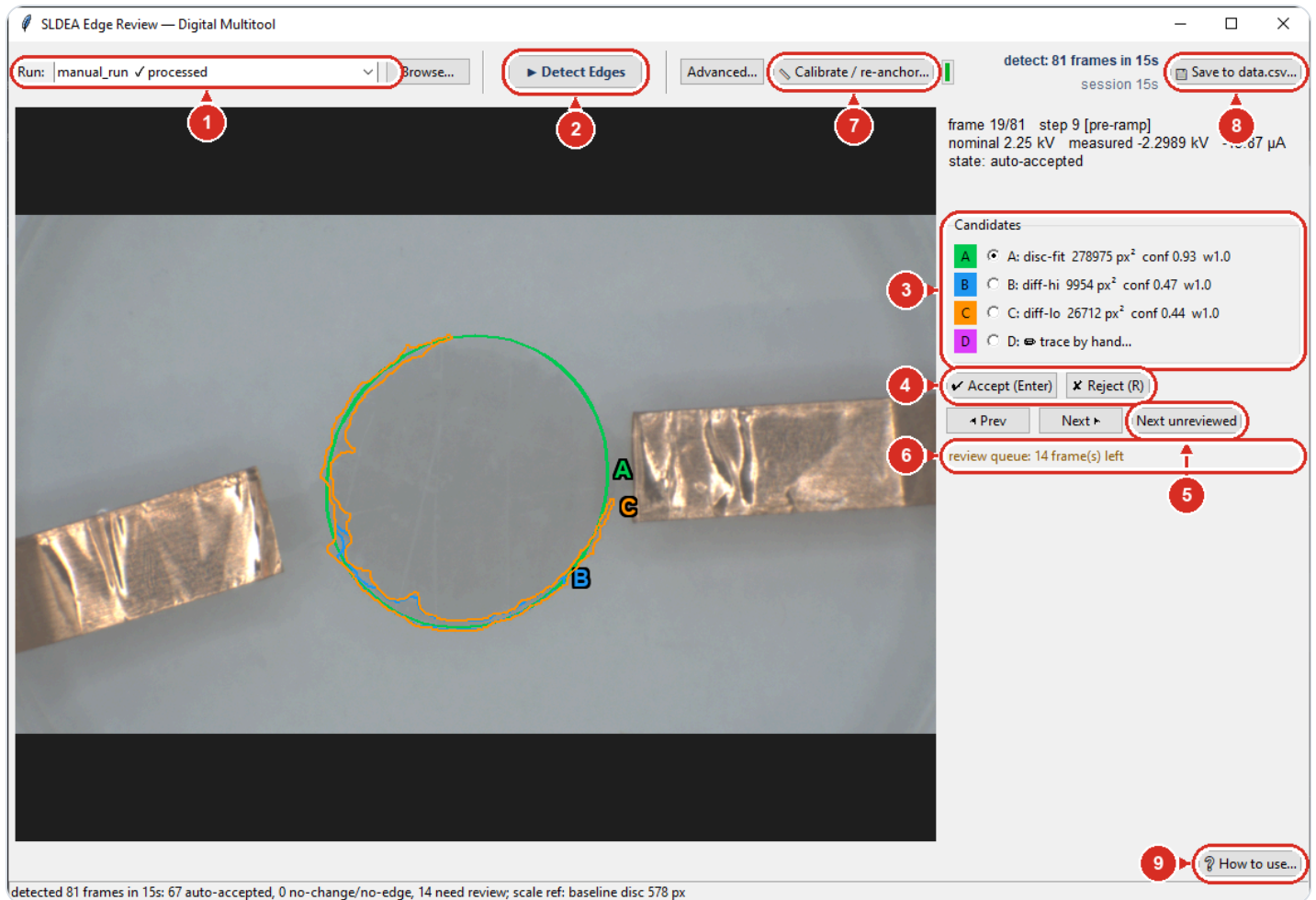
DIAGNOSTIC

When sliders can't fix a run, stop tuning and measure why: `Tune_SLDEA_Windows.bat /diag` writes `sldea_diag.txt/.json/.png` into the run folder and changes nothing. No window needed.

PLOT TOOL

Puts several finished runs on one figure: `python sldea_plot.py RUN1 RUN2 --mode area`. Writes a 300 dpi PNG *and* the tidy CSV behind it, so any figure can be traced back to its numbers. `--mode current` works on raw runs, before anyone has reviewed a frame.

Edge Review — reviewing a run



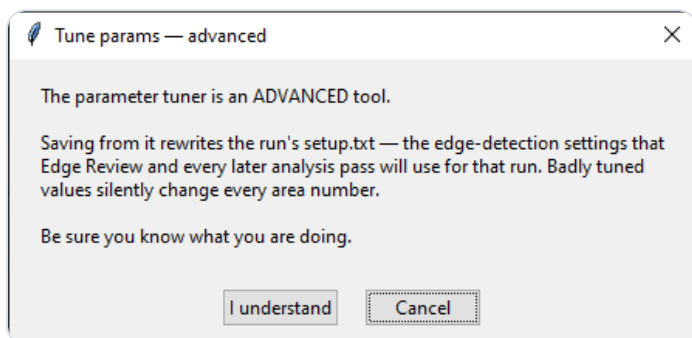
A real review frame (bench run P3_1 at 2.25 kV, pre-ramp): A = disc-fit (green), B/C = difference outlines. The machine's best pick is preselected — Enter agrees and moves on.

- 1 Pick the run — ✓ marks already-processed runs
- 2 Traces every frame — picks/rejects from earlier passes reset
- 3 Machine outlines — colors match the A/B/C tags on the image
- 4 Keys: 1/2/3 pick a candidate, Enter accepts, R rejects, D traces by hand
- 5 Jump to the next frame needing a human
- 6 Frames still waiting for a decision
- 7 Set or re-anchor the mm-per-px scale — verify the automatic disc fit
- 8 Writes accepted areas back (keeps a .bak) — confirms first
- 9 The review loop in one short read — start here

REVIEWING A RUN

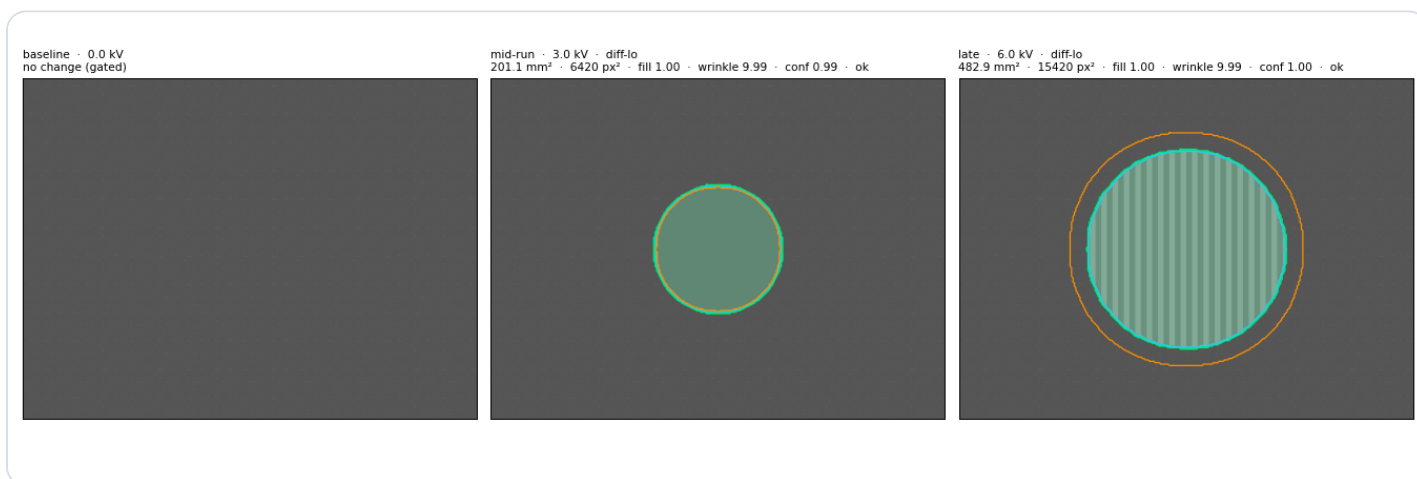
1. Open it from the SLDEA tab (🔍 Edge Review...) — or run `python sldea_edge_gui.py <run folder> --auto`.
2. Pick the run and click ► Detect Edges — a run with no scale anchor diverts to the  dialog first: ✓ Accept the machine's disc fit (or measure by hand if it refuses) and detection continues. Most frames auto-accept; the review queue holds the uncertain ones.
3. On each queued frame, pick the outline that hugs the electrode: keys 1/2/3 (A/B/C), R rejects, 4/D opens the hand tracer.
4. Enter accepts and moves on; Next unreviewed jumps ahead.
5. If areas look wrong by a constant factor,  Calibrate / re-anchor... — mid-review the corrected anchor is applied at Save; on a saved run it rewrites data.csv immediately, with no re-detection.
6. Queue empty →  Save to data.csv... — read the counts in the confirm prompt before answering Yes.

When outlines look wrong — tuning and diagnosis

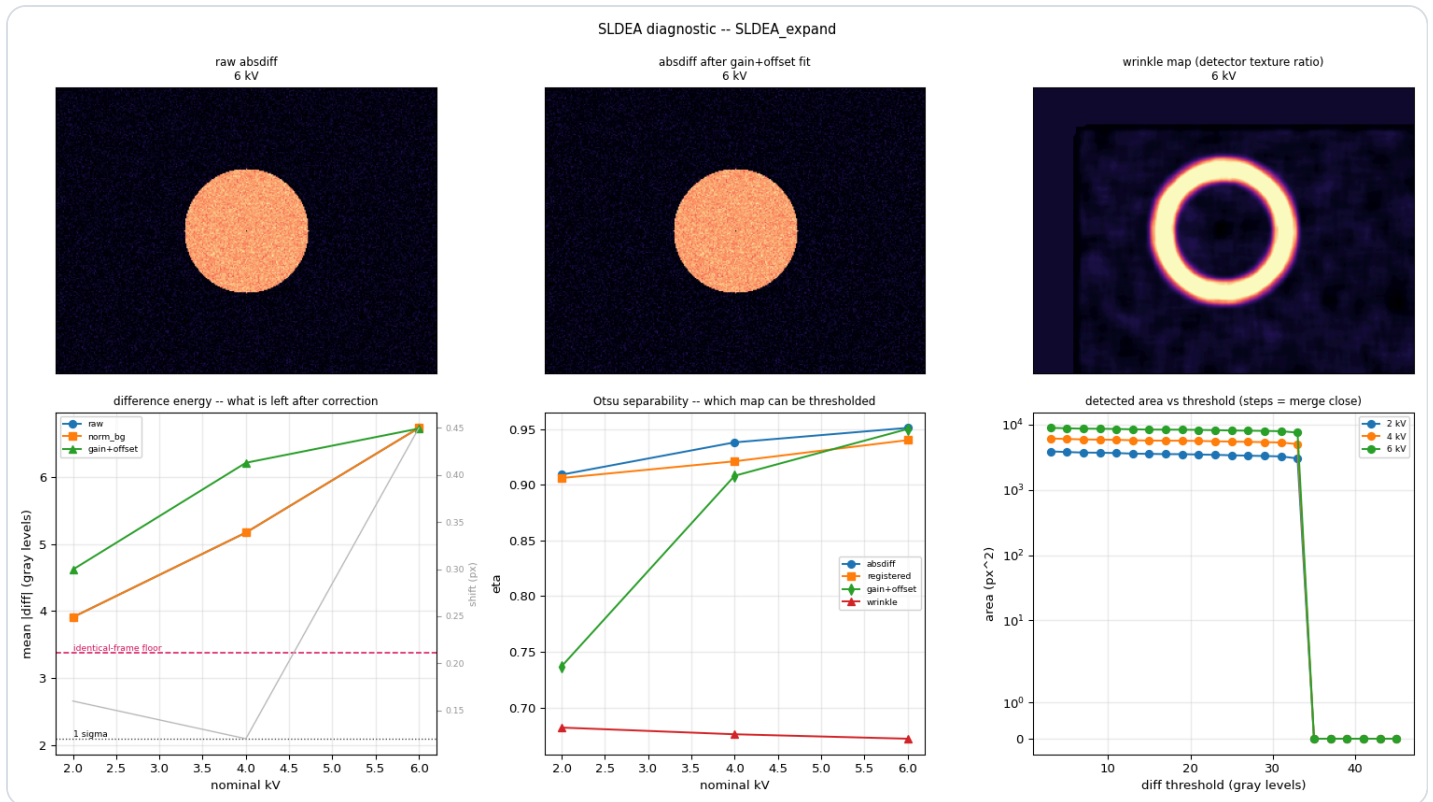


TUNE PARAMS IS AN ADVANCED TOOL

It asks you to confirm before it opens. Its **Save** rewrites the run's `setup.txt` — the detection settings Edge Review and every later analysis pass will use for that run — so badly tuned values silently change every area number. Reach for it only when Edge Review's outlines are consistently wrong across a run, and prefer per-frame picks or a hand trace for one-off bad frames.

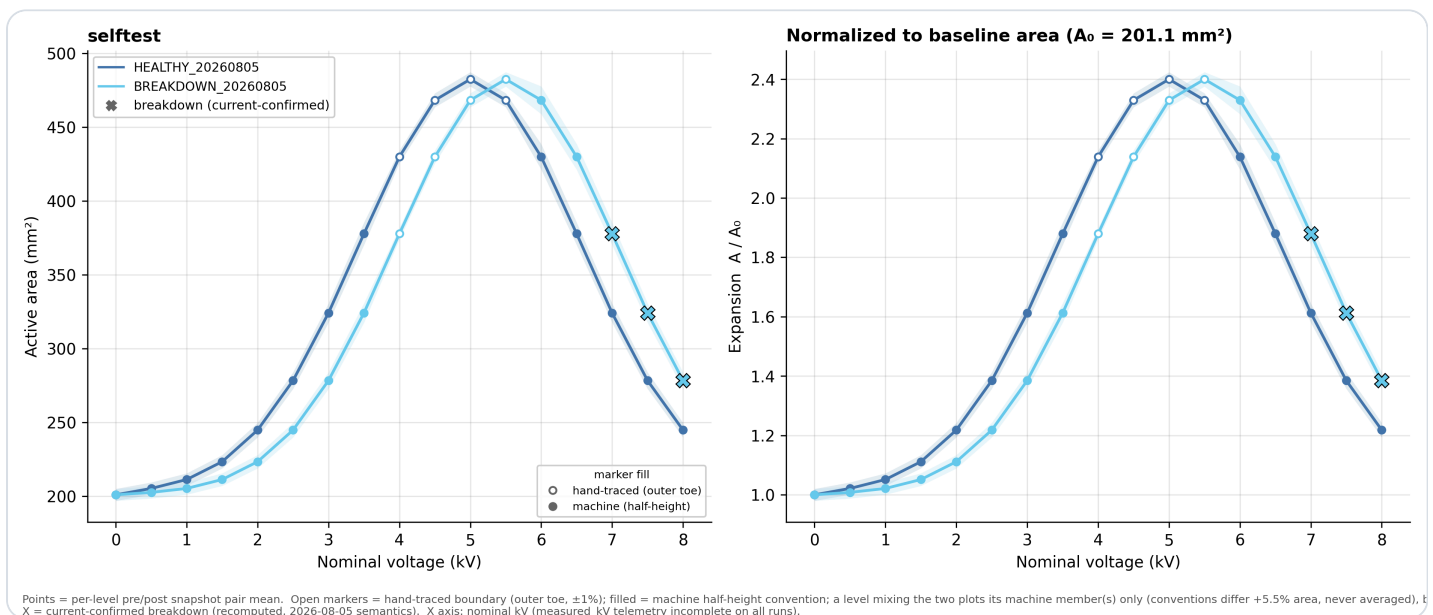


What the detector shows per frame: cyan = detected outline, orange = resting-disc reference. Confidence and area (mm²) are printed above each panel.



The diagnostic's verdict panels — run `--selftest` any time to check the stack without data.

Comparing finished runs — the plot tool



The plot tool's own self-test figure. Left: active area against voltage. Right: the same curves normalised to each run's baseline, so runs that started at different sizes can be compared. Shaded bands are the measurement uncertainty; open markers mean a hand-traced boundary; X marks a current-confirmed breakdown.

PUTTING RUNS ON ONE FIGURE

1. Review the runs first for `--mode area` — it plots measured areas, so it skips runs nobody has reviewed. `--mode current` and `--mode power` need no review at all.
2. `python sldea_plot.py RUN1 RUN2 --mode area --out ~/figures` — each RUN is a run folder, a parent full of runs, or a bench shortcut, exactly as for the tuner.
3. You get a PNG and a CSV of the same name. Keep them together: the CSV is the figure's evidence.
4. Breakdown $\times$ marks are recomputed from the recorded currents every time, never read off the frame filenames — a saved brand the currents do not support is reported as stale rather than drawn.

TYPICAL TUNING SESSION

1. Copy the run folder to local disk (tuning re-reads frames constantly; the share is slow).
2. Double-click Tune_SLDEA_Windows.bat, or drag the run folder onto it (bench runs: just type 1, 2 or 3).
3. Check the panel labelled baseline really shows the 0 kV image.
4. Drag sliders until the outlines hug the active area in all three panels.
5. Click  Save to setup.txt.
6. Open Edge Review on the same folder (`--auto` opens the scale-calibration dialog first — verify the machine's disc fit with $\checkmark$ Accept, or measure by hand if the fit refuses — and detection starts automatically once the anchor is in) and work the review queue.
7.  Save to data.csv... when the queue is empty; if it still will not tune, run the /diag diagnostic instead of another slider session.
8. Once two or more runs are reviewed, put them on one figure:  Plot runs... on the SLDEA tab (tick the runs, pick the mode, Export), or `sldea_plot.py RUN1 RUN2 --mode area` for the same figure from a script (`--mode current` works on raw runs, before any review).

WATCH OUT

Tuner Save writes into that run folder's setup.txt. If you tuned a local copy, copy setup.txt back to the master run or the review pass will not use your tuning.

If the run's baseline frame is missing or unreadable, the first loadable frame silently becomes the reference and every outline is wrong — always verify the baseline panel shows the 0 kV image.

In Edge Review,  Detect Edges starts clean: picks and rejects from a previous pass are discarded (hand-traced polygons are already safe in edge_labels.json). Save before re-detecting.

Edge Review's Save rewrites data.csv (a .bak is kept) and renames frame files from the first breakdown onward with a `_BREAKDOWN` suffix — read the counts in the confirm prompt before answering Yes.

▼ Shortcuts, environment and every tool control

Tune_SLDEA_Windows.bat	Double-click, or drag a run folder onto it; builds its own small Python environment on first run. Add /diag to run the diagnostic instead of the tuner.
Bench shortcuts 1 / 2 / 3	Typing 1, 2 or 3 as the run argument opens one of the three bench runs; only live on PCs where the OneDrive folder is synced, inert elsewhere.
SCPI_SLDEA_DIR	Set once (setx SCPI_SLDEA_DIR "C:\SLDEA", new window required) and every no-argument launch — launcher, tuner, tab buttons — resolves the newest run under it.
Blur (px, odd)	Tuner slider: noise smoothing before differencing — higher is smoother.
Diff thresh (0=auto Otsu)	Tuner slider: fixed change threshold; 0 lets the algorithm pick one per frame.
No-change gate (diff p99)	Tuner slider: below this the frame counts as unchanged vs baseline (no outline).
Min fill / Search ROI / Electrode mask / Wrinkle-mode sliders	The remaining shape and mask settings; the grey hint beside each slider says what it does.
Match frame to baseline: gain+offset (norm_bg)	Cancels camera exposure drift before differencing; leave on unless comparing to an old run.
 Save to setup.txt / Reset to defaults	Save writes the tuned values into the run's setup.txt for Edge Review and auto-process; Reset restores stock values.
Run: + Browse... + ► Detect Edges (Edge Review)	Pick a run (✓ processed marks finished ones) and trace every frame. Beside the progress bar sit two clocks: detect: freezes when the detection pass ends and keeps its frame count (81 frames in 2m14s), so runs stay comparable, while session counts how long the window has been open. A run you only re-anchor the scale on never detects, so its detect: line stays at not run.
Candidates A/B/C + D: trace by hand (Edge Review)	Pick the correct outline per queued frame — keys 1/2/3 pick, R rejects, 4/D opens the hand tracer, Enter accepts and moves on.

 Calibrate / re-anchor... (Edge Review)	Sets the run's px→mm anchor. Required per run (the scale gate): Detect and Save are blocked until it is done, and it overrides every automatic scale reference. The dialog opens on the machine's automatic disc fit for you to verify — ✓ Accept adopts it; measuring by hand (circle or two-point) is the fallback, offered with the fitter's own reason when the fit refuses. On a saved run with px measurements and no open review pass the same button re-anchors instead: every mm ² in data.csv is re-derived from the stored px and rewritten immediately, no re-detection needed — the dialog title says which of the two it is doing.
Advanced... (Edge Review)	Every detection setting with an explanation; Apply for this session, Apply + Save to write setup.txt. Re-run Detect afterwards.
 Save to data.csv... (Edge Review)	Writes accepted areas into the run's data.csv after a confirm prompt; keeps data.csv.bak and saves an area-vs-voltage plot plus outline overlays.
 How to use... (Edge Review)	Bottom-right of the window: the review loop in one short read — pick a run, Detect (calibration comes first, and the automatic disc fit is there to be verified, not repeated), work the queue, Save — plus the three things that cost people time: wash-out frames above ~5.5 kV are traced rather than rejected, an accepted scale is only staged until Save writes it, and a wrong scale can be re-anchored on a saved run without re-reviewing anything. It also says what conf and the w wrinkle index mean. Non-modal, so it can stay open beside the work.
sldea_diag.py (headless)	Run it on a run folder (or via /diag) when tuning fails; writes sldea_diag.txt/.json/.png into the run, changes nothing — the .json is small enough to send on.
sldea_plot.py (headless or windowed)	Puts one or more finished runs on a single figure. Takes the same run arguments as the tuner and the diagnostic (a run folder, a parent full of runs, or a bench shortcut) and writes a 300 dpi PNG plus the tidy per-snapshot CSV behind it, so a figure can always be traced back to its numbers. Colours are the house colourblind-safe set, assigned in the order runs are listed. Add --gui (or use  Plot runs... on the SLDEA tab) for the window: the same figure, with the runs and the toggles as tick boxes and a live preview. The command line stays the way to script a batch of figures.

--mode area | current | power

area (default) needs reviewed runs and draws active area against voltage plus the same curves normalised to each run's own baseline; current and power work on RAW runs too, so a run can be looked at electrically before anyone has reviewed a single frame. Breakdown crosses are always RECOMPUTED from the saved currents, never read off the frame names — an old area-based brand once marked 35 healthy frames.

Built from the live app (v1.2.0+0f65953) — every screenshot is a real capture, every callout is anchored to the actual control. Sources: `README.md`, `SLDEA_HANDOFF.md`, `SLDEA_MEASUREMENT.md` and the code itself.